

8/13 TeV NNLO theory predictions: content summary

Summary of the theory-prediction grid stored in `withVVF_data_nlc.json` (`ttbarj-nnlo-cms-atlas-analysis` submodule, built by its `convertData.py`). Restricted to the ρ -related differential observables, inclusive cross sections, parameters and PDF variations; non- ρ differential distributions are omitted.

Grid dimensions

- **Energies:** 8 TeV and 13 TeV.
- **Orders in the converted JSON:** LO, NLO, NNLO (default NNLO prediction) and NNLO_noVVF (the older VVF-subtracted variant), at both energies. NNLO and NNLO_noVVF differ in every bin.
- **Theory nuisance parameters:** per scale point and PDF family, `tnp_eigenvectors` holds per-bin TNP shifts (Chebychev0 model on the series [LO, NLO–LO, NNLO–NLO]). NNLO has 10 eigenvectors (the 10th is the subleading-colour term), NNLO_noVVF has 9. `normalized_expanded` holds the expanded-ratio scale variation.
- **Top-quark mass grid:** 167.5, 171.5, 172.5, 173.5, 177.5 GeV, for every order and energy.
- **Scale variations:** 3 central scale choices (H_T , $H_T/2$, $H_T/4$), each with the standard 7-point (μ_R, μ_F) envelope, consolidated onto 13 distinct (μ_R, μ_F) points per mass/order. Nominal central scale is $H_T/2$.
- **Inclusive cross section:** ϵ -expansion coefficients [$c_0, c_{-1}, c_{-2}, c_{-3}, c_{-4}$] from the STRIPPER dimensional-regularization output; c_0 (pb) is the physical cross section, $c_{-1} \dots c_{-4}$ are IR-pole coefficients kept as a numerical cancellation check.

ρ -related differential distributions

Label in JSON	Source key	Binning	Description
<code>rhocms8</code> / <code>rhocms13</code>	<code>rhocms</code>	1 binning, 4 bins	ρ with CMS-style selection, energy-tagged
<code>rhoatlas8</code>	<code>rhoatl1</code> (binning 1)	8 bins	ρ with ATLAS-8 TeV-analysis binning
<code>rhoatlas13</code>	<code>rhoatl1</code> (binning 2)	7 bins	ρ with ATLAS-13 TeV-analysis binning

`rhoatlas8` and `rhoatlas13` are stored at *both* theory energies (the “8”/“13” denotes the ATLAS binning scheme, not the collision energy); `rhocms` is energy-tagged instead. ρ is the top-mass-sensitive jet observable of Alioli, Fernandez, Fuster, Irles, Moch, Uwer, Vos, *A new observable to measure the top-quark mass at hadron colliders*, Eur. Phys. J. C 73 (2013) 2438, arXiv:1303.6415.

PDF sets

Eigenvector/error-set members are only fully populated at the physical central scale ($H_T/2$); off-diagonal scale points carry only the central (member 0) PDF for each family. LHAPDF IDs and member counts were cross-checked against LHAPDF’s official `pdfsets.index` and match the

Family (JSON key)	LHAPDF ID	Members stored	Role	Reference
PDF4LHC21_40	93100	41 (0–40: central + 40 eigenvectors)	PDF eigenvector set, all scale points	PDF4LHC Working Group, J. Phys. G 49 (2022), arXiv:2203.05506
CT18NNLO	14000	59 (0–58: central + 29 eigenvector pairs)	PDF eigenvector set	Hou et al. (CT18NNLO), Phys. Rev. D 103 (2021), arXiv:1912.01005
CT18NNLO_as_0117	14067	1	α_s down-variation of CT18NNLO	same
CT18NNLO_as_0119	14069	1	α_s up-variation of CT18NNLO	same
MSHT20nnlo_as118	27400	65 (0–64: central + 32 eigenvector pairs)	PDF eigenvector set	Bailey, Cridge, Lang, Martin, Thorne, Phys. J. C 81 (2021), arXiv:2012.04684
MSHT20nnlo_as_smallrange	27500	members 3, 4 only	α_s variation members (fine grid around 0.118)	same
NNPDF31_nnlo_as_0118_hessian	304400	101 (0–100: central + 100 Hessian eigenvectors)	PDF eigenvector set	NNPDF Coll. (B1), Eur. Phys. J. C 77 (2017), arXiv:1706.00428
NNPDF31_nnlo_as_0117	323100	1	α_s down-variation	same
NNPDF31_nnlo_as_0119	323500	1	α_s up-variation	same
ABMP16als118_5_nnlo	42780	30 (0–29: central + 29 eigenvectors)	PDF eigenvector set (nf=5)	Alekhin, Blümlein, Placakyte, Phys. Rev. D 96, 014011 (2017), arXiv:1701.05838
ABMP16als117_5_nnlo	42750	1	α_s down-variation	same
ABMP16als119_5_nnlo	42810	1	α_s up-variation	same

member counts present in this repository’s converted data; the literature references are the standard citations for these PDF groups.